Do Presidents Govern as They Speak?

Measuring the Say-Do Gap in U.S. Presidential Texts

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Abstract

This project studies whether U.S. presidents govern consistently with the ideological positions they express in public language. I separate presidential text into two corpora: a Say corpus, composed of interviews, speeches, inaugural addresses, farewell addresses, news conferences, and related public-facing texts, and a Do corpus, composed of executive orders and written presidential orders. The raw documents were collected from The American Presidency Project. The core idea is to place each president twice inside a two-dimensional ideological space: once based on what the president says, and once based on what the president formally does. The first dimension measures left-right economic ideology, while the second measures libertarian-authoritarian ideology. I use a theory-driven dictionary-based text-as-data method. Because presidential documents vary substantially in length, words are used as the main measuring device for corpus balance. The final scoring method uses document-level phrase presence, phrase caps, family caps, and evidence filtering to avoid over-weighting repeated legal or administrative language. Presidents with insufficient Say or Do evidence are filtered out before final comparison. The Say-Do Gap is measured as the Euclidean distance between each president's Say point and Do point.

1 Introduction

Presidents often campaign, speak, and govern through language. They explain values in speeches, defend priorities in interviews, and communicate political identity through public-facing rhetoric. However, presidents also govern through formal written action. This creates a central empirical question: do presidents govern in the same ideological direction that they speak?

This project measures the gap between presidential rhetoric and presidential action using text as data. For each president, I build one ideological profile from public rhetoric and another ideological profile from formal executive action. I then compare these two profiles inside the same two-dimensional ideological space.

The project does not attempt to produce a moral ranking of presidents. Instead, it builds a reproducible measurement framework. The goal is to estimate how closely presidential words and presidential actions align when both are measured using the same ideological dictionary, the same scoring rules, and the same evidence filters.

2 Research Question

The central research question is:

> Do U.S. presidents govern consistently with the ideological positions they express publicly?

More specifically, this project asks whether there is a measurable distance between a president's rhetorical ideology and governing ideology. Rhetorical ideology refers to the ideological position expressed through public-facing language, such as interviews, speeches, inaugural addresses, farewell addresses, and news conferences. Governing ideology refers to the ideological position reflected in executive orders, which are formal instruments of presidential action.

The project asks five related questions:

1. Where does each president fall ideologically based on public rhetoric?
2. Where does each president fall ideologically based on formal executive action?
3. How large is the geometric distance between the two positions?
4. Which presidents appear most and least aligned between words and actions?
5. Which party appears most and least aligned between words and actions?

The main outcome is the Say-Do Gap. A smaller gap means the president's rhetoric and executive action are closer in ideological space. A larger gap means the president's rhetoric and executive action are farther apart.

3 Data and Corpus Construction

The project uses two main presidential text corpora: a Say corpus and a Do corpus. These are built from raw presidential documents collected from The American Presidency Project at the University of California, Santa Barbara [1]. The Say corpus captures public-facing presidential language, while the Do corpus captures formal executive action.

3.1 The Say Corpus

The Say corpus represents what presidents say publicly. It includes presidential interviews, speeches, inaugural addresses, farewell addresses, news conferences, and public letters or statements when available. These documents are useful because they capture how presidents explain their values, justify their decisions, and present their ideological identity to the public.

3.2 The Do Corpus

The Do corpus represents what presidents formally do. It includes executive orders and written presidential orders. These documents are important because they are not only symbolic statements. They are formal governing texts that direct agencies, create administrative priorities, regulate behavior, and exercise presidential authority.

This distinction is the foundation of the project. The Say corpus captures presidential rhetoric. The Do corpus captures presidential action.

3.3 Raw Corpus Composition

Figure 1 shows the raw word count by document type. This confirms that the corpus is not evenly distributed across document categories. News conferences, executive orders, and interviews provide most of the available text, while inaugural addresses and written presidential orders are much smaller sources.

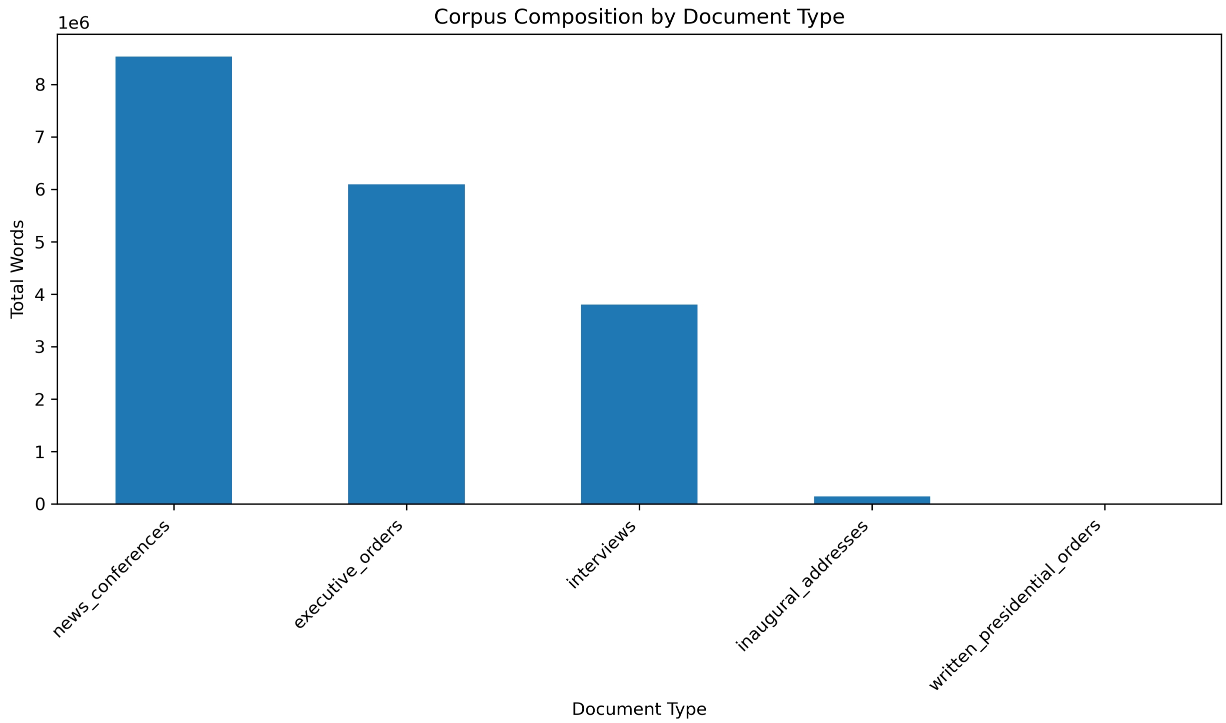

Figure 1: Raw corpus composition by document type, measured using total words.

The project uses words as the main measurement unit because the documents are very different from each other and have very different lengths. A single news conference can contain thousands of words, while a written presidential order can be much shorter. Therefore, counting documents would create a misleading comparison. Measuring total words gives a more realistic view of how much language is available for analysis.

3.4 Say Versus Do Corpus Size

Figure 2 summarizes the full Say and Do corpora. The Say corpus contains 12,484,484 words, while the Do corpus contains 6,107,851 words. The Say corpus is therefore larger than the Do corpus.

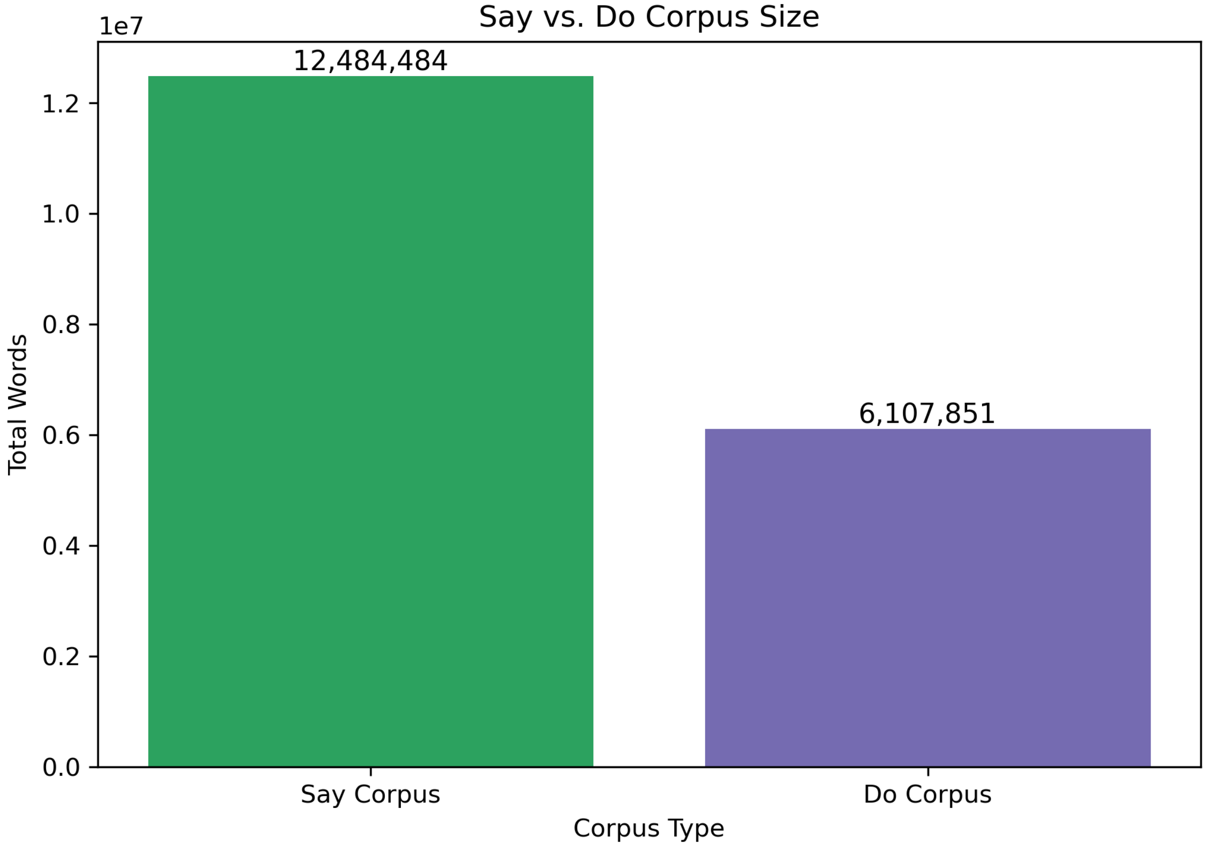

Figure 2: Total Say versus Do corpus size, measured using total words.

This imbalance was expected. Spoken presidential language is more common than formal action execution. Presidents give speeches, answer questions, hold news conferences, and make public remarks more frequently than they issue executive orders or written presidential orders. This does not make the comparison invalid, but it makes normalization necessary. For that reason, the project does not compare raw ideological hit counts. Instead, all ideological evidence is normalized per 1,000 words.

3.5 Corpus Balance by President

Figure 3 shows the share of each president's available corpus that comes from Say and Do documents. This matters because the project compares each president's rhetoric against formal action. Some presidents have a balanced mix of Say and Do text, while others are dominated by one corpus type.

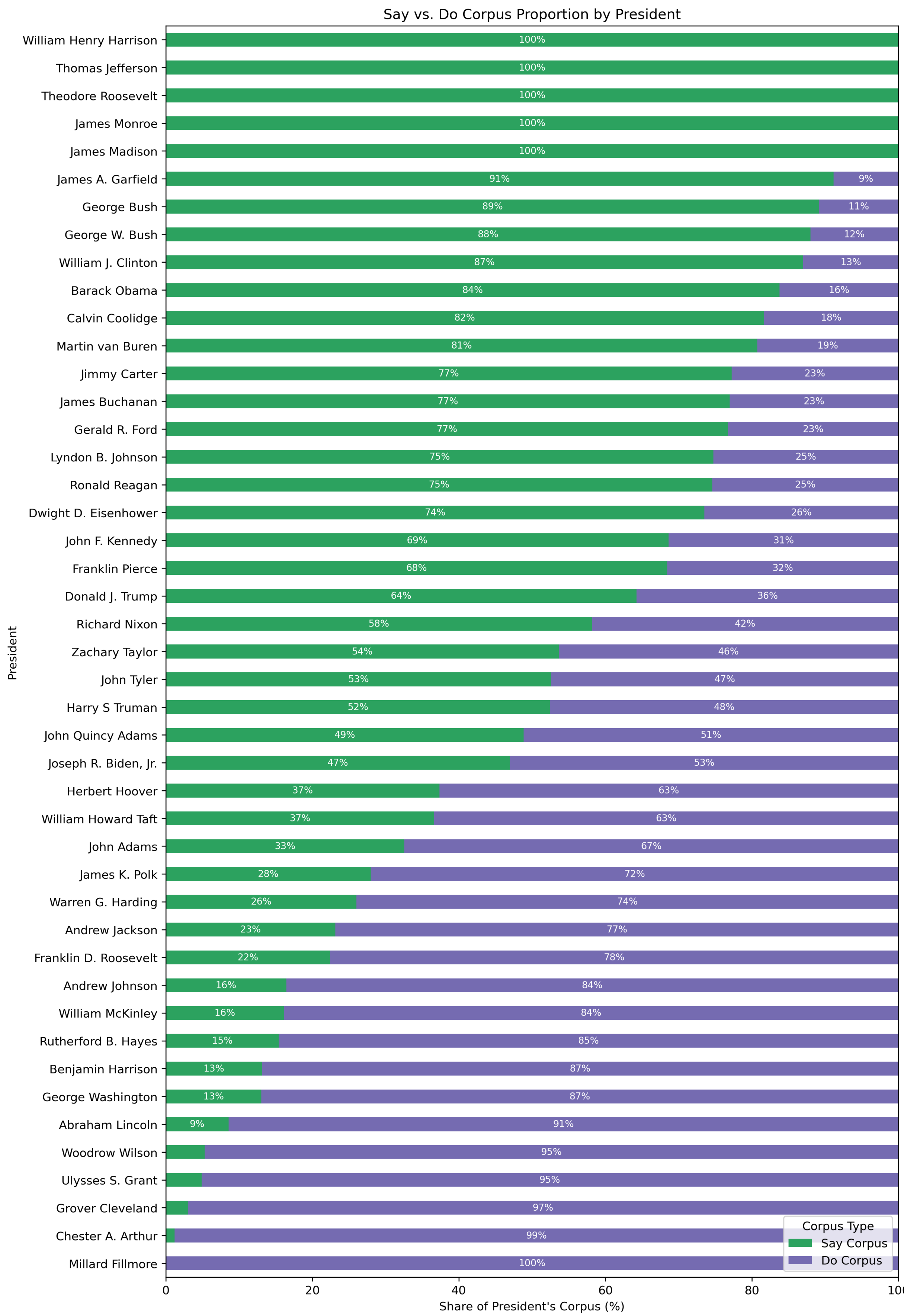

Figure 3: Say versus Do corpus proportion by president.

The president-level imbalance reinforces the need for evidence filtering. A president should not be treated as highly aligned or highly divergent if one side of the comparison is based on very little text.

3.6 Timeline Coverage

Figure 4 shows the historical coverage of the Say and Do corpora by decade. The corpus is much stronger in the modern period, especially from the twentieth century onward. Earlier presidents generally have less available text, especially for public-facing Say documents.

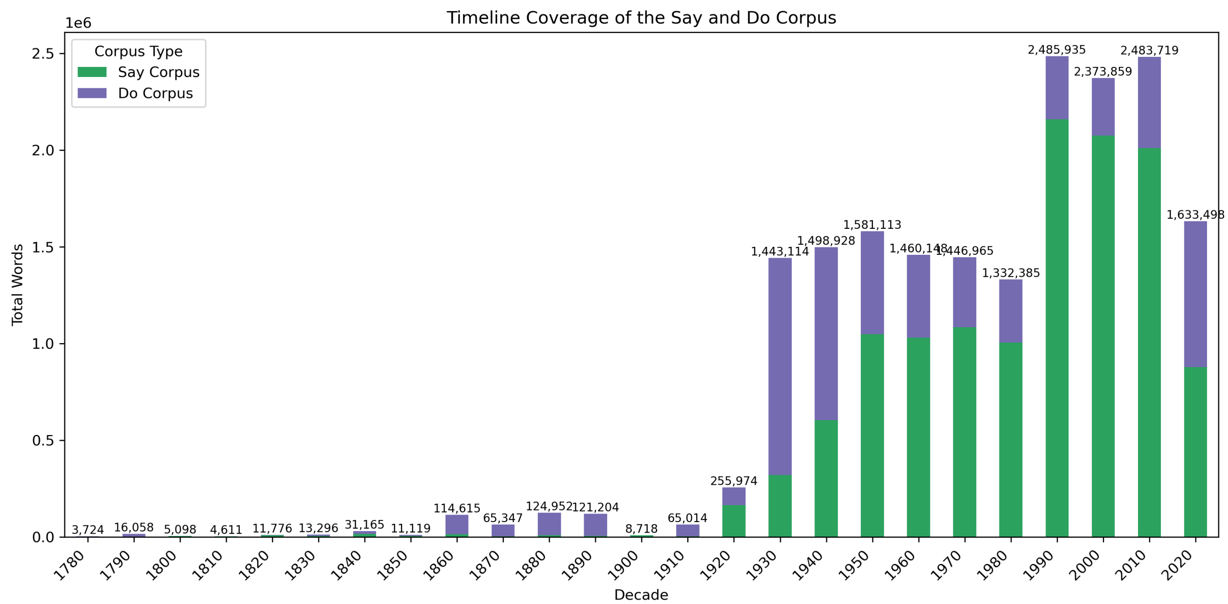

Figure 4: Timeline coverage of the Say and Do corpora by decade, measured using total words.

This historical imbalance is important because the project compares presidents across a long time period. The later corpus is richer, while earlier presidents often have less surviving or scraped text. For that reason, the final pipeline includes minimum evidence rules before presidents are included in the main comparison.

3.7 Document Length Distribution

Figure 5 shows the distribution of document lengths by document type. This figure is central to the corpus design because it shows that presidential documents are not equal-sized units. Interviews, news conferences, and inaugural addresses are generally much longer than executive orders and written presidential orders.

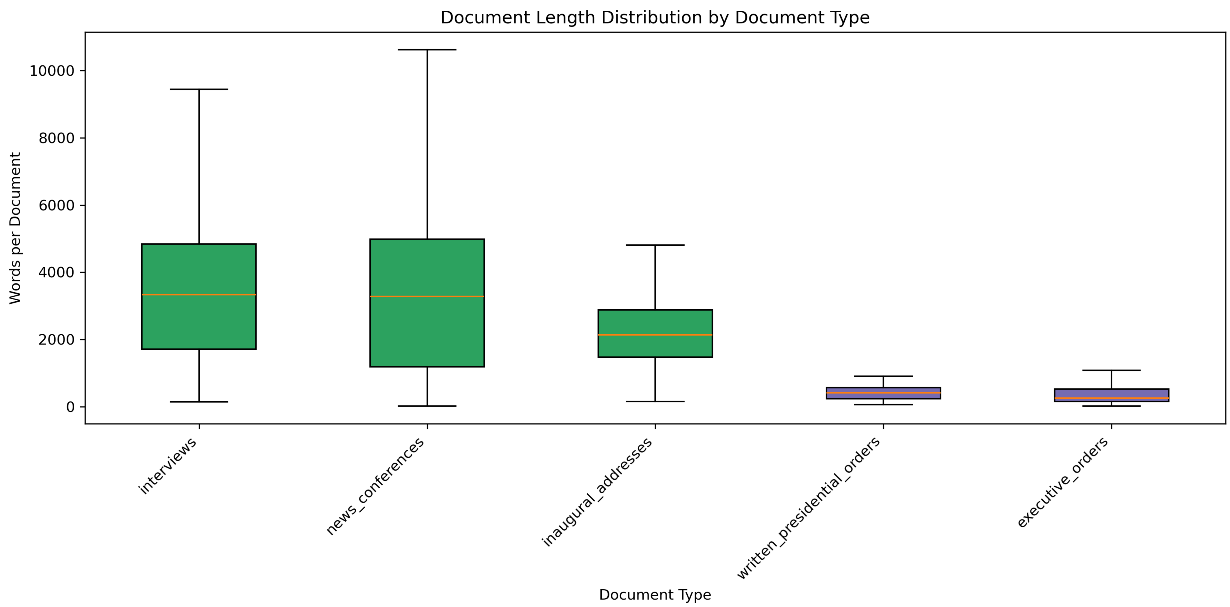

Figure 5: Document length distribution by document type, measured as words per document.

This is the strongest visual justification for using words as the unit of measurement. If the project counted documents, a short written presidential order and a long news conference would receive the same weight. That would be misleading because they contain very different amounts of language. Word counts provide a more stable and comparable unit across document types.

3.8 Document Structure

The scraped files are organized using a two-tab structure. One tab contains document-level metadata, including a document identifier, president, date, title, and source information. The second tab contains chunked text, where each document is split into smaller pieces for processing. The document identifier connects each chunk back to the metadata table.

The raw storage unit is the chunk. However, the main analytical unit is the president-corpus pair. This means that all Say text for a president is combined into one Say profile, and all Do text for a president is combined into one Do profile.

President	Corpus Type	Text Group
Franklin D. Roosevelt	Say	Speeches / interviews / addresses
Franklin D. Roosevelt	Do	Executive orders / written orders
Ronald Reagan	Say	Speeches / interviews / addresses
Ronald Reagan	Do	Executive orders / written orders

This structure allows a direct comparison between what each president says and what each president does.

3.9 Why Words Are the Measuring Device

Because the documents in the corpus vary substantially in length, the project uses words as the main measuring device for corpus size and balance. A document-level count would give the same weight to a short speech and a long executive order, even though they contain very different amounts of text. A chunk-level count would also depend on the technical chunking process rather than the substantive amount of language available for analysis.

Word counts provide a more stable and comparable unit across interviews, speeches, inaugural addresses, farewell addresses, news conferences, executive orders, and written presidential orders. Therefore, corpus balance is evaluated primarily using total words, not total documents or total chunks.

This decision also affects the scoring method. Dictionary hits are normalized as hits per 1,000 words so that presidents with larger corpora are not automatically treated as more ideological simply because they have more text.

3.10 Excluded Presidents Due to Low Coverage

Not every president had enough usable Say and Do evidence after preprocessing. To avoid misleading comparisons, presidents with low coverage were excluded from the final Say-Do distance calculation. These presidents remain part of the raw corpus description, but they are not used in the final alignment ranking.

Table 1 lists the excluded presidents ordered by presidential period.

Table 1: Presidents excluded from the final Say-Do comparison because of low Say coverage, low Do coverage, or weak usable dictionary evidence.

Period	President	Exclusion Reason
1789–1797	George Washington	Low Say coverage
1797–1801	John Adams	Low Say and Do coverage
1801–1809	Thomas Jefferson	Low Say and Do coverage
1809–1817	James Madison	Low Say and Do coverage
1817–1825	James Monroe	Low Do coverage
1825–1829	John Quincy Adams	Low Say and Do coverage
1829–1837	Andrew Jackson	Low Say coverage
1837–1841	Martin Van Buren	Low Say and Do coverage
1841	William Henry Harrison	Low Do coverage
1841–1845	John Tyler	Low Say and Do coverage
1845–1849	James K. Polk	Low Say coverage
1849–1850	Zachary Taylor	Low Say and Do coverage
1850–1853	Millard Fillmore	Low Say and Do coverage
1853–1857	Franklin Pierce	Low Say and Do coverage
1857–1861	James Buchanan	Low Say and Do coverage
1861–1865	Abraham Lincoln	Low Say coverage
1865–1869	Andrew Johnson	Weak usable dictionary evidence
1869–1877	Ulysses S. Grant	Low Say coverage
1877–1881	Rutherford B. Hayes	Low Say coverage
1881	James A. Garfield	Low Say and Do coverage
1881–1885	Chester A. Arthur	Low Say coverage
1885–1889; 1893–1897	Grover Cleveland	Low Say coverage
1889–1893	Benjamin Harrison	Low Say coverage
1901–1909	Theodore Roosevelt	Low Say and Do coverage
1909–1913	William Howard Taft	Weak usable dictionary evidence
1913–1921	Woodrow Wilson	Low Say coverage
1921–1923	Warren G. Harding	Low Say coverage
1923–1929	Calvin Coolidge	Weak usable dictionary evidence

The exclusion rule is important because the project compares two points for each president: one Say point and one Do point. If either side has too little text, the resulting distance could look artificially small or artificially large. Filtering therefore protects the validity of the final Say-Do comparison.

4 Preprocessing

Preprocessing was designed to prepare the text for ideological scoring without becoming the full methodology. The goal was to reduce technical noise while preserving substantive meaning. I did not want preprocessing to decide ideology. Instead, preprocessing creates a cleaner and more comparable text base for the later scoring stage.

4.1 Full-Document Reconstruction Before Preprocessing

Before preprocessing, the corpus-building logic was corrected so that chunked rows are reassembled into full documents before analysis. This matters because phrase-based scoring should not happen on artificial chunks. A politically meaningful phrase could be split across `chunk_no` boundaries and lose meaning if the model scores each chunk separately.

This connects to a core text-as-data idea: the document choice matters. The unit of analysis should reflect the substantive object being studied, not a storage artifact created for scraping or processing convenience. In this project, chunks are useful for storage, but full documents are better for preprocessing and phrase recovery.

4.2 Basic Text Cleaning

The first preprocessing step removed formatting noise. This included URLs, line breaks, repeated spaces, HTML artifacts, curly quotation marks, long dashes, and non-standard punctuation. The purpose was not to change meaning. The purpose was to make phrase matching more consistent. Cleaning removes noise and improves tractability, while normalization makes case, punctuation, and formatting more consistent.

4.3 Executive-Order Boilerplate Removal

The next preprocessing step removed legal and genre-specific boilerplate from the Do corpus. This included repeated executive-order phrases such as "by the authority vested in me," "it is hereby ordered," "pursuant to," standard closing clauses, and repeated legal or administrative framing language.

This was necessary because executive orders naturally contain authority-heavy language. If left untreated, this boilerplate could falsely inflate the authoritarian or state-control score. In other words, the model might measure the legal genre of executive orders rather than the president's substantive ideological position.

This step addresses measurement confounding. The text measure should capture the hidden concept of interest, which is ideology, not merely the formal style of executive-order writing.

4.4 Negation Standardization

The next step standardized contractions. Examples include “don't” becoming “do not,” “can't” becoming “cannot,” and “won't” becoming “will not.” At this stage, negation was not interpreted. The goal was only to prepare the text so that later scoring could detect context more reliably.

4.5 Normalization and Confidence Filtering

The final preprocessing step created the final preprocessed text column and counted final words. These word counts are important because later ideological scores are calculated per 1,000 words. This connects to the term-frequency idea: the relevant measure is not only how many times a word or phrase appears, but how often it appears relative to the total number of tokens.

The project also filtered presidents based on whether they had enough Say and Do text after preprocessing. This prevents presidents with very little evidence from looking falsely aligned or falsely extreme.

5 Methodology

5.1 Methodological Overview

The final project used a dictionary-based Text-as-Data pipeline to estimate the gap between presidential rhetoric and presidential action. The method began by separating the corpus into two conceptually different text collections: the Say corpus and the Do corpus. The Say corpus represented public presidential language, while the Do corpus represented formal governing action through executive orders and written presidential orders. Both corpora were preprocessed before scoring so that dictionary matching would be more consistent across document types.

The main measurement tool was a four-pole ideological dictionary. The dictionary used two ideological axes: left-right and libertarian-authoritarian. Each axis had two directions, producing four poles: left, right, libertarian, and authoritarian. Each dictionary entry contained a phrase family, axis, direction, intensity score, theme, phrase variant, cleaned phrase variant, and direction sign. This gave the project a fully explicit and auditable scoring structure. Every phrase that contributed to a president's score could be traced back to a specific ideological family and axis.

5.2 From N-Grams to Dictionary Scoring

The first exploratory methodology step used 2-grams, 3-grams, and 4-grams to discover common phrases in the Say and Do corpora. This helped reveal repeated language patterns. However, the most frequent phrases were often procedural or source-specific artifacts, such as legal boilerplate, institutional titles, and common presidential wording.

Therefore, n-grams were useful for exploration but not strong enough as the final scoring method. The final methodology uses dictionary-based ideological scoring because the project requires theory-driven ideological measurement, not simply frequent phrase extraction.

5.3 Building the Four-Pole Ideological Dictionary

This stage builds the project's final four-pole ideological dictionary used to score each president's Say corpus and Do corpus. The dictionary is fully explicit in the scoring script and contains two ideological axes: left-right and libertarian-authoritarian. Each axis has two directions, producing four poles: left, right, libertarian, and authoritarian.

Each dictionary family is assigned an axis, direction, intensity level from 1 to 5, theme, and phrase variants. Intensity 1 represents weak or broad ideological language, while intensity 5 represents extreme ideological language. For example, the left pole ranges from public services and worker support to public ownership and revolutionary socialism. The right pole ranges from small business and business confidence to radical laissez-faire and corporate rule. The libertarian pole ranges from individual liberty and local self-government to anarchy and stateless society. The authoritarian pole ranges from public order and emergency language to dictatorship, political repression, and ethnonationalist authoritarianism.

The script expands family-level entries into a phrase-level dictionary where each variant becomes one row. Text variants are normalized, duplicates are audited, directional signs are assigned, and the final dictionary is exported for the scoring model. This produces a transparent, reproducible measurement instrument for mapping presidential rhetoric and executive action onto a two-dimensional ideological space.

5.4 Final Sample and Evidence Filtering

The scoring procedure loaded the final dictionary, the preprocessed Say and Do corpora, and the president evidence filter table. Only presidents that passed the confidence filter were considered. Additional exclusions were then applied to remove presidents with historically weak or incomparable evidence. The final sample was restricted to presidents with enough usable Say and Do text to support a meaningful comparison.

The scoring block also applied a minimum evidence threshold. A president needed at least 25 raw document-level dictionary hits in both the Say corpus and the Do corpus. Presidents below this threshold were not included in the final ranking because their ideological coordinates would be too sensitive to a small number of phrase matches.

This filtering step is important because a dictionary score is only meaningful when enough evidence exists in both corpora. Without this rule, a president with only a handful of phrase hits could appear artificially extreme, artificially moderate, falsely aligned, or falsely divergent.

5.5 Dictionary-Based Ideological Scoring

After building the final four-pole ideological dictionary, the next stage applied the dictionary to the preprocessed Say corpus and Do corpus in order to estimate each president's rhetorical and governing position in ideological space. The main goal of this stage was to place each president twice in the same ideological coordinate system: once based on public language and once based on governing action. The Say point represents the president's public rhetoric, while the Do point represents the president's executive orders and written presidential actions. The distance between these two points is the project's main Say-Do Gap measure.

For each president-corpus pair, the scoring procedure searched the text using one optimized compiled regular expression containing all dictionary phrase variants. In particular, executive orders contain repeated legal and administrative phrases that can dominate raw dictionary counts. One phrase repeated many times in the Do corpus could overwhelm the entire ideological score for a president. To correct this, the final scoring method moved from raw phrase counts to a more conservative document-level phrase-presence method.

5.6 Document-Level Phrase Presence

A key methodological correction was changing the scoring from raw phrase repetition to document-level phrase presence. A dictionary phrase is counted at most once per document. This means that if a phrase appears many times in the same executive order, it does not repeatedly inflate the president's score. This reduced the effect of boilerplate repetition and made the Do corpus less vulnerable to artificial ideological inflation.

The project uses phrase matching, not sentiment scoring. The text is normalized, and each dictionary variant is searched in the president's Say and Do text. The matching process uses longest-match logic and avoids overlapping phrase matches. This prevents a long phrase such as "protect against surveillance" from also counting "surveillance" separately inside the same phrase.

5.7 Variant and Family Caps

The scoring layer also added caps to prevent any one phrase or phrase family from overwhelming the results. Each dictionary variant was capped at a maximum number of document-level hits per president-corpus pair, and each phrase family was also capped. Specifically, each dictionary variant was capped at 25 document-level hits per president-corpus, and each phrase family was capped at 100 hits per president-corpus.

This was necessary because validation showed that a single phrase, such as a broad administrative or economic phrase, could otherwise drive a president's entire ideological placement. The final score therefore reflects a broader pattern of phrase-family evidence rather than uncontrolled repetition.

5.8 Ideological Space

Each president is placed in a two-dimensional ideological matrix.

The x-axis measures left-right economic ideology:

- -5 = global left extreme,
- 0 = center,
- $+5$ = global right extreme.

The y-axis measures libertarian-authoritarian ideology:

- -5 = libertarian extreme,
- 0 = center,
- $+5$ = authoritarian extreme.

The scale is global, not only American. U.S. presidents are expected to cluster closer to the center because mainstream U.S. presidents are usually not communist, anarchist, fascist, or dictatorial in global ideological terms.

5.9 Dictionary Structure: Phrase Families

The final method uses a large ideological phrase dictionary organized by phrase families. This matters because different phrases can express the same political idea. For example, "support local business," "support small business," "small business owners," and "entrepreneurship" all point toward a similar economic theme. Instead of treating these as unrelated terms, the dictionary groups them into one family, such as `support_local_enterprise`.

Each dictionary entry contains the phrase family, ideological axis, ideological direction, intensity score, theme, phrase variant, cleaned phrase variant, and direction sign. This makes the dictionary auditable. After scoring, the project can inspect which phrase families drove a president's score. Appendix A reports the dictionary families used in the final model.

5.10 Intensity Scores and Direction Signs

Each phrase family receives an intensity score from 1 to 5:

Intensity	Meaning
1	Weak or broad ideological signal
2	Moderate-low ideological signal
3	Clear ideological signal
4	Strong ideological signal
5	Extreme ideological signal

This avoids treating every phrase equally. For example, "support small business" is weaker than "privatize everything," and "public order" is weaker than "dictatorial rule." The updated dictionary uses intensity 5 for global ideological extremes, such as revolutionary socialism, anarchism, anarcho-capitalism, dictatorship, suspension of democracy, political repression, and ethnonationalist authoritarianism.

Each family is also assigned an ideological direction. For the left-right axis, left equals -1 and right equals $+1$. For the libertarian-authoritarian axis, libertarian equals -1 and authoritarian equals $+1$.

Each phrase hit becomes a signed intensity:

$$SignedIntensity = DirectionSign \times Intensity \tag{1}$$

A left phrase with intensity 3 becomes -3. An authoritarian phrase with intensity 4 becomes $+4$. A libertarian phrase with intensity 4 becomes -4.

5.11 Axis Score Calculation

For each president and corpus type, phrase hits were aggregated by variant, family, axis, and direction. Each hit was weighted by its dictionary intensity score. Direction signs were then applied: left and libertarian phrases received negative signs, while right and authoritarian phrases received positive signs.

The final axis score was calculated as signed weighted hits divided by total axis hits. For president p, corpus type c, and axis a, the axis score is:

$$AxisScore_{p,c,a} = \frac{\sum SignedWeightedHits_{p,c,a}}{\sum DictionaryHits_{p,c,a}} \tag{2}$$

This produced one left-right score and one libertarian-authoritarian score for each president in each corpus. The president's final coordinate is therefore:

$$x = LeftRightScore \tag{3}$$

$$y = LibertarianAuthoritarianScore \tag{4}$$

5.12 Say-Do Gap Construction

Each president was then placed twice in the ideological space. The Say point used the president's Say corpus scores. The Do point used the president's Do corpus scores. The horizontal coordinate represented the left-right axis, and the vertical coordinate represented the libertarian-authoritarian axis.

The project then calculated the Say-Do Gap as the Euclidean distance between the Say point and the Do point:

$$SayDoGap_p = \sqrt{(x_{Say,p} - x_{Do,p})^2 + (y_{Say,p} - y_{Do,p})^2} \tag{5}$$

Larger distances indicate larger separation between rhetoric and action, while smaller distances indicate greater alignment.

5.13 Validation and Audit Process

The scoring process created validation outputs to audit the results. These included variant-level hit tables, family-level hit tables, long-format axis scores, president-corpus scores, and Say-Do gap rankings. These outputs made it possible to inspect which phrase families were driving each president's score.

The validation step was important because it revealed global scoring problems, including generic executive-order language, repeated phrase domination, and low-evidence presidents. Each of these issues was then corrected in the scoring layer.

The validation process identified three global issues:

1. repeated phrase domination could distort scores;
2. generic legal or administrative language could burden the Do corpus;
3. low-evidence presidents could produce unstable scores.

Repeated phrase domination was fixed with document-level presence counting and caps. Generic executive-order language was addressed through the Do-safe dictionary revision and scoring caps. Low-evidence presidents were handled through manual exclusions and the 25-hit minimum threshold.

After these corrections, the results became more politically defensible. The Say corpus map generally reflects rhetorical ideology, while the Do corpus map reflects a more conservative estimate of governing ideology. The Do corpus no longer mechanically pushes all presidents toward authoritarianism, which was a major improvement.

5.14 Final Methodological Interpretation

The final output should be interpreted as a dictionary-based ideological proxy, not as a definitive expert ranking of presidential ideology. The purpose is not to claim that the dictionary perfectly captures ideology, but to create a transparent, reproducible measure of the distance between what presidents say and what presidents do.

The final methodology is stronger than the initial version because it corrects for three major sources of bias: executive-order boilerplate, phrase repetition, and low-evidence presidents. The resulting Say-Do Gap is therefore more defensible as an exploratory Text-as-Data measure. The project can now use the final filtered sample, document-level dictionary scoring, capped phrase-family contributions, and Euclidean distance as the main quantitative framework for comparing presidential rhetoric and presidential action.

6 Results

6.1 Overall Say-Do Gap

The average Say-Do Gap among the selected presidents is 2.01. This means that, on average, presidents' rhetorical positions and executive-action positions are separated by about two ideological units in the two-dimensional matrix. Because the axes range from -5 to $+5$, this average suggests a meaningful but not extreme distance between what presidents say and what they do.

Figure 6 visualizes the Say-Do movement for each selected president. Each arrow connects a president's Say position to the corresponding Do position. This allows the project to show not only the size of the gap, but also the direction of movement between rhetoric and executive action.

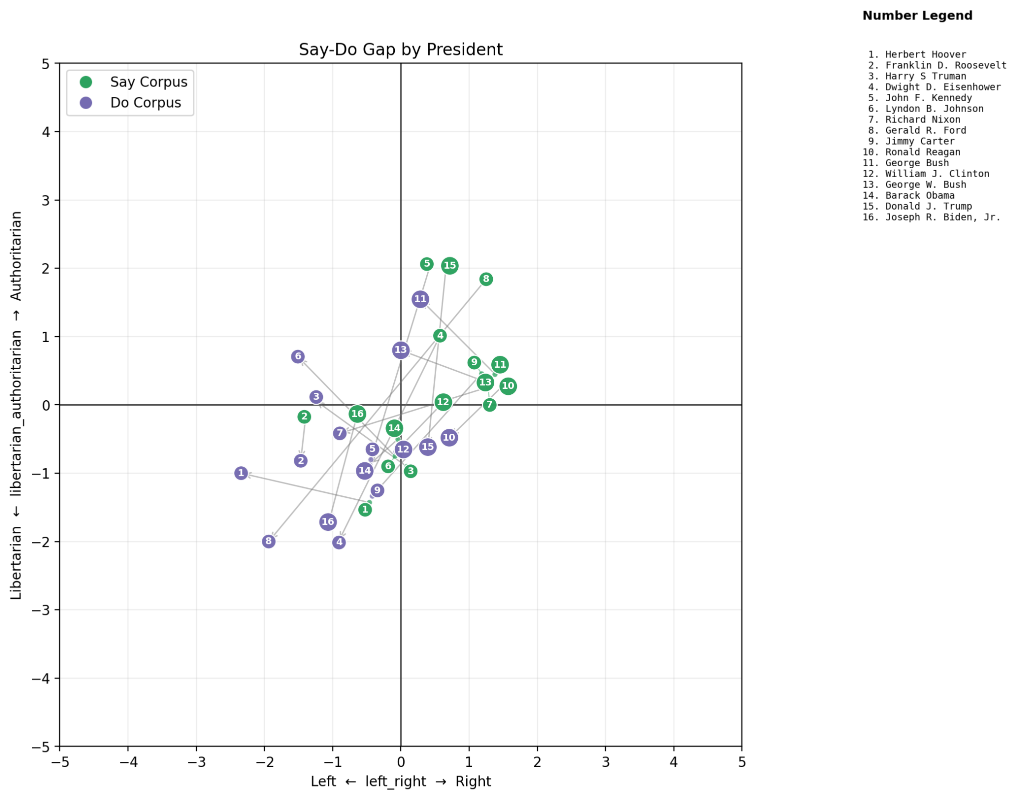

Figure 6: Say-Do Gap by president. Arrows connect each president's rhetorical position to the corresponding executive-action position.

Figure 7 ranks presidents by the size of their Say-Do Gap. This makes the main outcome easier to interpret because it directly shows which presidents have the largest and smallest measured distance between rhetoric and formal executive action.

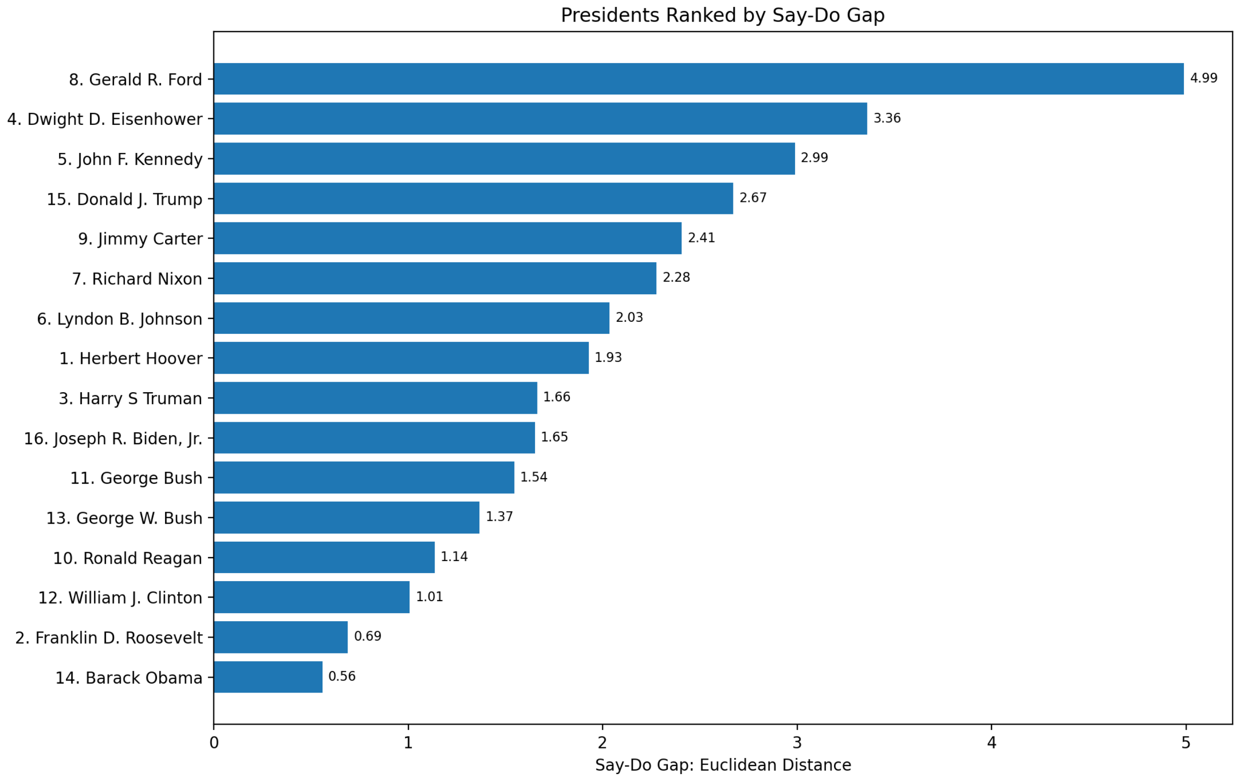

Figure 7: Ranked Say-Do Gap by president. Larger values indicate greater distance between rhetoric and executive action.

Visually, most president-level points are contained within the $[-2, 2]$ range on both axes. This clustering is substantively important. The ideological dictionary was built on a global scale, where the extremes include revolutionary socialism, anarchism, anarcho-capitalism, dictatorial rule, political repression, and ethnonationalist authoritarianism. Against that global ideological range, U.S. presidential politics appears more contained and less extreme. The visual compression of most presidents toward the center therefore should not be interpreted as a failure of the measurement strategy. Instead, it suggests that American presidential politics is less ideologically diverse than the full global political spectrum represented by the dictionary.

6.2 Interpretation of the Ideological Maps

The full-scale maps are useful for showing that U.S. presidents generally do not occupy the extreme corners of the ideological space. Most presidents fall closer to the center than to the poles. This is consistent with the expectation that mainstream U.S. presidents operate within a narrower ideological range than the full global spectrum.

6.3 President-Level Dictionary-Linked Interpretation

Table 2 provides a short dictionary-linked interpretation for each selected president. These readings should be interpreted as diagnostic summaries of the phrase families that appear to drive each president's Say-Do distance. They are not independent historical rankings. Instead, they connect the quantitative gaps back to the dictionary families that produced the scores.

Table 2: President-level dictionary-linked interpretation of Say-Do gaps.

President	Short dictionary-linked read
Gerald R. Ford	Largest gap; driven by `privacy_from_state`, `due_process_protections`, and `military_strength` terms such as "right to privacy," "due process," and "strong military."
Dwight D. Eisenhower	Large gap; authority-axis difference tied to `dictatorial_rule`, `privacy_from_state`, and `tax_reduction` language.
John F. Kennedy	Large gap; pulled by `military_strength`, `dictatorial_rule`, and `voluntary_civil_society` terms.
Donald J. Trump	Large gap; driven by `anti_institutional_populism` and `free_speech_press`, with terms such as "deep state," "fake news," and "free press."
Jimmy Carter	Meaningful gap; tied to `free_speech_press`, `military_strength`, `political_repression`, and `climate_intervention`.
Richard Nixon	Notable gap; driven by `military_strength`, `death_penalty_punitive`, `climate_intervention`, and `voluntary_civil_society`.
Lyndon B. Johnson	Medium-large gap; linked to `law_and_order`, `free_speech_press`, and `assembly_protest_rights`.
Herbert Hoover	Economic gap; driven by `public_ownership`, `public_investment`, and `local_self_government` terms.
Harry S Truman	Medium gap; likely split between `military_strength`, `public_investment`, and civil-liberty families.
Joseph R. Biden, Jr.	Medium gap; likely driven by `climate_intervention`, `public_investment`, `free_speech_press`, and authority-restraint terms.
George Bush	Smaller-medium gap; likely tied to `military_strength`, `nationalist_sovereignty`, and `free_market/tax_reduction` language.
George W. Bush	Smaller-medium gap; likely tied to `military_strength`, `nationalist_sovereignty`, `law_and_order`, and national-security language such as `national_security_basic` and `surveillance_security`.
Ronald Reagan	Smaller gap; rhetoric and action are closer, with both pulled by `tax_reduction`, `free_market`, `military_strength`, and `small_government`.
William J. Clinton	Low gap; relatively aligned around `welfare_reform_work_requirements`, `free_trade_globalization`, and moderate market/state language.
Franklin D. Roosevelt	Very low gap; Say and Do align around `public_investment`, `labor_union_power`, `social_safety_net`, and `public_ownership`.
Barack Obama	Smallest gap; strong alignment around `public_health_state_capacity`, `climate_intervention`, civil-liberties families, and `public_investment`.

The table shows that the largest gaps are not random. They are usually tied to identifiable phrase families on the authority axis, the economic axis, or both. Ford, Eisenhower, Kennedy, and Trump have larger gaps because their Say and Do corpora are pulled by different combinations of liberty, military, institutional, and authority-related families. By contrast, Obama, Franklin

D. Roosevelt, Clinton, and Reagan show smaller gaps because their rhetoric and action are more consistently pulled by similar families on the same axes.

6.4 Historical Interpretation

Some individual president positions should be interpreted in historical context. Herbert Hoover is a useful example. Hoover's Say-Do Gap and ideological placement may be affected by the Great Depression. His presidency occurred during an extreme economic crisis, and crisis conditions can change both presidential rhetoric and formal executive action. As a result, Hoover's measured distance may reflect not only his ideological commitments, but also the unusual governing pressures produced by the Great Depression.

This point highlights a broader interpretive rule for the project: the Say-Do Gap should not be read as a simple measure of hypocrisy or inconsistency. A large gap may reflect strategic rhetoric, institutional constraints, crisis conditions, changing political coalitions, or differences between public persuasion and formal executive authority.

7 Party-Level Aggregation

After president-level scoring, the project aggregates results by party. The party aggregation does not rerun the dictionary. Instead, it uses the already-created president-level family-hit output. This prevents excluded presidents from re-entering the analysis at the party level.

Only presidents who passed the final evidence check are included in party aggregation. Parties are mapped into Democratic, Republican, and Other. The final two-party analysis keeps only Democratic and Republican presidents and excludes Other.

Party Say-Do distance is calculated using the same Euclidean distance formula used for presidents. Figure 8 shows the party-level movement from Say to Do positions. This provides a compact view of whether party-level rhetoric and executive action move in similar or different ideological directions.

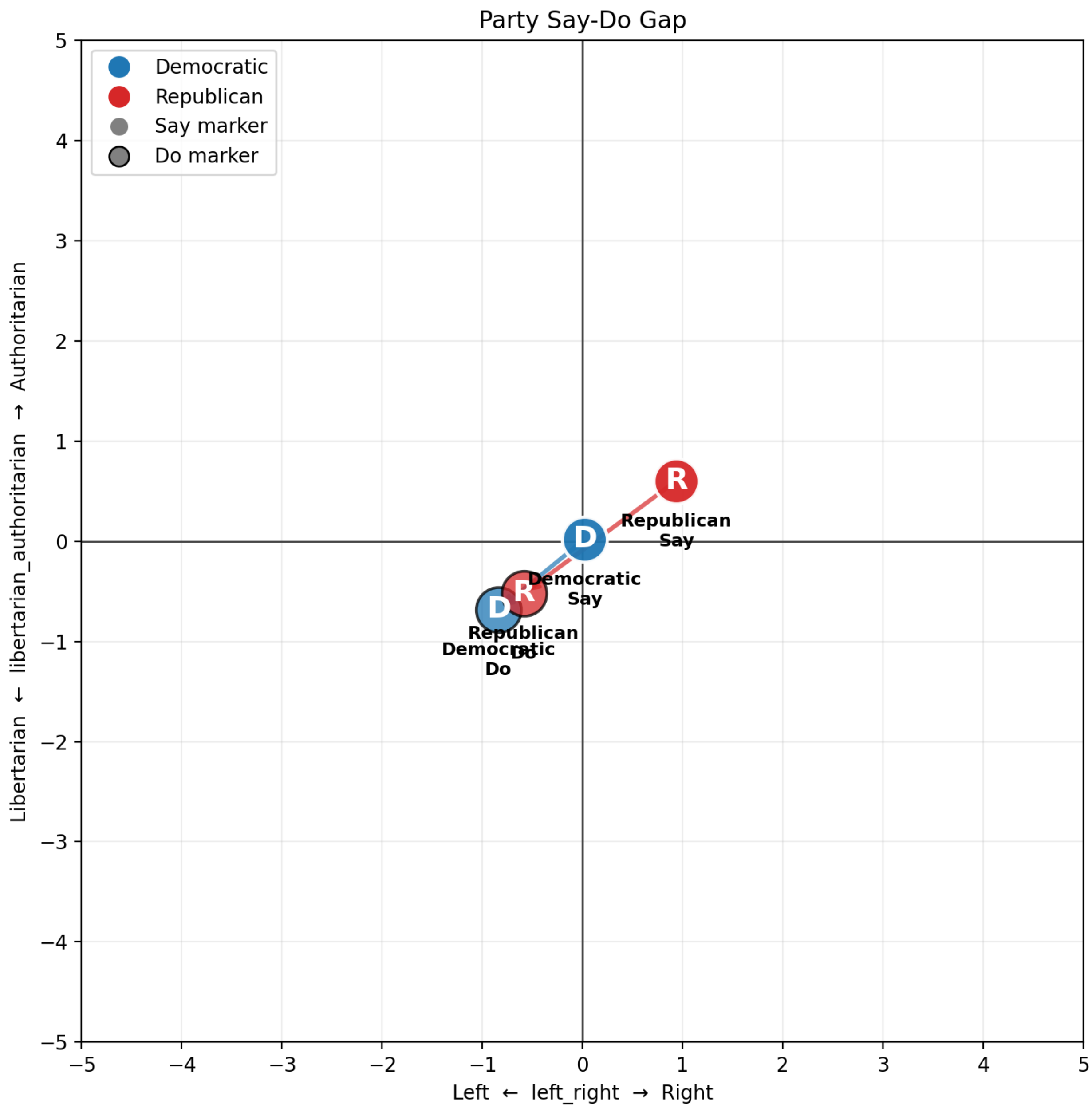

Figure 8: Party-level Say-Do Gap. Arrows connect each party's rhetorical position to its executive-action position.

Figure 9 ranks the party-level Say-Do distances. This provides a simplified comparison of which party shows a larger aggregate gap between public rhetoric and formal executive action in the filtered sample.

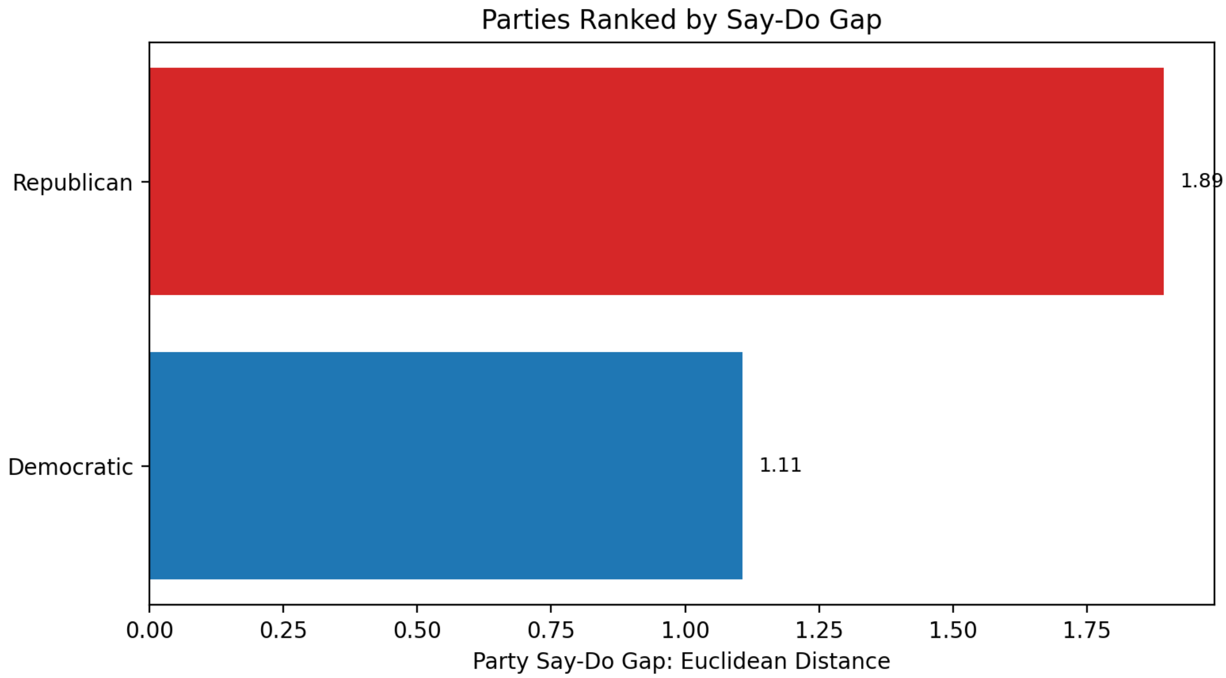

Figure 9: Ranked party-level Say-Do Gap. Larger values indicate greater aggregate distance between party rhetoric and party executive action.

8 Limitations

1. **Dictionary and hyperparameter risk.** Dictionary-based measurement creates researcher-defined parameter risk. The ideological dictionary, phrase families, directions, and intensity weights are not discovered automatically from the data. They are defined by the researcher. In this project, the dictionary design was supported by an LLM-assisted synthesis of political categories and expert-style ideological distinctions, but it was not validated through an independent survey of political scientists. This creates hyperparameter risk: different reasonable dictionary choices could produce different scores.

2. **Historical instability of ideological language.** Ideological labels are historically unstable. What counts as left, right, libertarian, or authoritarian can change across time. A phrase that signals one ideological position in the nineteenth century may signal something different in the twentieth or twenty-first century.

3. **Anachronism.** By design, an exercise like this has some level of anachronism. The project places presidents from very different historical periods into one shared ideological space. This makes comparison possible, but it also risks applying modern or global ideological categories to presidents who governed in very different institutional, economic, and social contexts.

4. **Era-specific political meaning.** The ideological axes may reflect the politics of each era more than a universal scale. Federal power, civil rights, labor policy, national security, market regulation, and executive authority did not mean exactly the same thing under Lincoln, Franklin D. Roosevelt, Reagan, Obama, Trump, and Biden.

5. **Executive-order genre effects.** Executive orders have a distinctive legal and administrative style. They often contain words related to authority, enforcement, direction, compliance, and regulation because of the document type itself. This can make the text appear more authoritarian even when the substantive policy is not authoritarian.

6. **Executive-power-only design.** This project focuses on executive power as the operational measure of governing action. That is a useful and observable form of presidential action, but it is not the only way the country is governed. Presidents also shape policy through legislation, appointments, budgets, military decisions, agency rulemaking, judicial nominations, diplomacy, party leadership, and informal agenda-setting. The Supreme Court, Congress, federal agencies, state governments, and other institutional actors also shape governing outcomes. Therefore, the Do corpus captures one important channel of presidential power, but it does not rank or measure the Supreme Court, Congress, or other branches and institutions of power.

7. **Limited contextual understanding.** A dictionary method cannot fully understand context. The model can count terms such as "security," "rights," or "regulation," but it may not know whether the president is expanding, limiting, defending, or criticizing those concepts.

8. **Coverage tradeoff.** The evidence filter improves reliability but reduces coverage. Some presidents may be excluded not because they are unimportant, but because the available Say or Do evidence is too limited for a fair comparison.

Because of these limitations, the results should be interpreted as a baseline measurement, not as a final truth about presidential ideology.

9 Answer to the Research Question

We defined our research question as "Do U.S. Presidents govern consistently with the ideological positions they express publicly?" The answer to this question, as expected, is yes, as there are difficulties in governing a country. The more interesting part is the way this difference is measured and compared across generations.

A small Say-Do Gap suggests that a president's public rhetoric and formal action are ideologically aligned. A large Say-Do Gap suggests greater divergence between what the president says and what the president formally does.

10 Conclusion

This project builds a text-as-data framework for comparing presidential rhetoric and presidential action. The final methodology uses full-document reconstruction, focused preprocessing, theory-driven ideological calibration, phrase-family matching, document-level phrase presence, intensity weights, word-count normalization, phrase caps, family caps, audit tables, and evidence-based president filtering.

The final result shows an average Say-Do Gap of 2.01 among the selected presidents. Most presidential points remain visually concentrated near the center of the global ideological space, especially inside the $[-2, 2]$ region. This suggests that U.S. presidential politics is more ideologically contained than the global ideological extremes represented by the dictionary. The project therefore works best as a comparative measure inside the American presidency, not as a claim that U.S. presidents occupy the full global ideological spectrum.

The results should also be interpreted with caution because some president-level placements are open to historical debate. Conventional historical perception might place Richard Nixon as more authoritarian because of his law-and-order politics, and George W. Bush as more authoritarian because of the expansion of national security policy after September 11. Herbert Hoover may also be expected to appear farther to the economic right, but his corpus is likely affected by the extraordinary context of the Great Depression, which changed both the language and governing priorities of his presidency. These cases show that the model captures textual evidence from the available corpus, not a complete expert judgment of each president's historical ideology.

At the same time, some results align with broad historical expectations. Ronald Reagan appears relatively strong on the economic right, while Franklin D. Roosevelt appears relatively strong on the economic left. These placements provide face-validity support for the dictionary approach, even though individual coordinates should still be interpreted as approximate text-based estimates rather than definitive ideological classifications.

The final output is a two-dimensional ideological position for each president's Say corpus and Do corpus. The distance between these points becomes the Say-Do Gap. This framework makes it possible to compare presidents and parties in a transparent, reproducible way.

A Dictionary Audit Table

This appendix reports the final four-pole ideological phrase-family dictionary used in the model. In the full scoring script, each family contains multiple phrase variants. The table reports each family, ideological axis, direction, intensity, theme, and representative examples. The table is formatted in portrait orientation so it remains readable without horizontal pages.

Table 3: Final four-pole dictionary families used for ideological scoring.

Family	Axis	Direction	Int.	Theme	Representative Variants
local public goods	left-right	left	1	weak public investment	public service; community services; public facilities; public resources; common good
basic worker support	left-right	left	1	weak worker support	working families; working people; ordinary workers; protect workers; dignity of work
basic consumer protection	left-right	left	1	weak market regulation	protect consumers; consumer protection; fair prices; fair competition; market fairness
public investment	left-right	left	2	state investment	public investment; infrastructure investment; federal investment; public works; community development
education public spending	left-right	left	2	public education	public education; school funding; student aid; pell grants; universal pre-k
public health state capacity	left-right	left	2	public health	public health; disease prevention; health care access; community health centers; rural healthcare

Family	Axis	Direction	Int.	Theme	Representative Variants
housing tenant support	left-right	left	2	housing affordability	affordable housing; rental assistance; tenant protections; housing vouchers; fair housing
care economy public support	left-right	left	2	care state support	child care assistance; affordable child care; elder care support; paid sick leave; care economy
social safety net	left-right	left	3	welfare state	social safety net; protect social security; food assistance; unemployment insurance; protect medicare
labor union power	left-right	left	3	labor power	collective bargaining; right to organize; union rights; workplace protections; union jobs
living wage	left-right	left	3	labor income	living wage; minimum wage; paid family leave; equal pay; good paying union jobs
economic inequality	left-right	left	3	economic equality	income inequality; wealth inequality; economic justice; working class; new deal
anti poverty policy	left-right	left	3	anti-poverty	war on poverty; reduce poverty; economic opportunity; head start; low income assistance
market regulation	left-right	left	3	economic regulation	regulate corporations; antitrust enforcement; monopoly power; wall street reform; big pharma
climate intervention	left-right	left	3	climate state action	climate action; clean energy; renewable energy; reduce emissions; environmental protection
redistribution tax wealth	left-right	left	4	economic redistribution	tax the wealthy; wealth tax; pay their fair share; progressive taxation; corporate minimum tax
universal healthcare	left-right	left	4	healthcare state role	universal healthcare; medicare for all; public option; health care as a right; public health insurance
democratic socialism	left-right	left	4	democratic socialist policy	democratic socialism; social democracy; worker ownership; public banking; guaranteed income
public ownership	left-right	left	5	state ownership	nationalize; public ownership; state owned enterprise; nationalize energy; government ownership
revolutionary left	left-right	left	5	revolutionary socialism	abolish capitalism; class struggle; workers revolution; seize the means of production; collectivization
support local enterprise	left-right	right	1	weak small business market	support small business; small business owners; local business; entrepreneurship; family owned business
business confidence	left-right	right	1	weak business support	business confidence; business community; job creators; support employers; encourage investment
free trade globalization	left-right	right	2	market opening	free trade; open markets; trade liberalization; expand trade; global markets
economic nationalism	left-right	right	2	economic nationalism	protect american jobs; tariffs; made in america; buy american; domestic production
energy independence fossil	left-right	right	2	energy market right	energy independence; oil and gas; coal industry; fossil fuels; pipeline construction

Family	Axis	Direction	Int.	Theme	Representative Variants
welfare reform work requirements	left-right	right	2	market workfare	welfare reform; work requirements; personal responsibility; reduce dependency; self sufficiency
school choice market education	left-right	right	2	market education	school choice; charter schools; education vouchers; parental choice; education savings accounts
faith based private delivery	left-right	right	2	private social services	faith based organizations; charitable choice; private charities; voluntary organizations; church based
rugged individualism voluntaryism	left-right	right	2	voluntary market conservatism	rugged individualism; self reliance; voluntary action; private charity; local responsibility
tax reduction	left-right	right	3	tax market right	tax cuts; lower taxes; tax relief; capital gains tax cut; lower tax rates
supply side growth	left-right	right	3	supply side	supply side economics; pro growth policies; capital formation; business investment; investment tax credit
deregulation	left-right	right	3	market freedom	deregulation; cut red tape; regulatory burden; regulatory relief; excessive regulation
free market	left-right	right	3	market liberalism	free market; free enterprise; economic freedom; private sector; market incentives
small government	left-right	right	3	limited government	limited government; smaller government; federal overreach; government interference; reduce bureaucracy
fiscal austerity	left-right	right	3	fiscal conservatism	balanced budget; reduce spending; spending cuts; fiscal discipline; reduce the deficit
property rights	left-right	right	3	property market right	property rights; private property; property owners; home ownership; ownership society
privatization	left-right	right	4	privatization	privatize; privatization; private sector solution; market based solution; private delivery
radical laissez faire	left-right	right	4	minimal state market	laissez faire; minimal state; abolish welfare; abolish regulation; minimal government
anarcho capitalist market extreme	left-right	right	5	market extreme	abolish the state; privatize everything; private police; taxation is theft; stateless capitalism
oligarchic corporate rule	left-right	right	5	corporate extreme	corporate rule; rule by corporations; corporate sovereignty; corporate state; market rule over democracy
individual liberty basic	libertarian-authoritarian	libertarian	1	weak individual liberty	individual liberty; personal freedom; individual rights; personal liberty; freedom of conscience
voluntary civil society	libertarian-authoritarian	libertarian	1	voluntary society	voluntary association; civil society; private association; mutual aid; free association
local self government	libertarian-authoritarian	libertarian	1	local autonomy	local control; local self government; local decision making; local autonomy; decentralized authority
government restraint	libertarian-authoritarian	libertarian	2	limited state power	restrain government; limit government power; checks on power; constitutional limits; government by consent

Family	Axis	Direction	Int.	Theme	Representative Variants
privacy from state	libertarian-authoritarian	libertarian	2	privacy rights	privacy rights; personal privacy; protect privacy; privacy from government; digital privacy
due process protections	libertarian-authoritarian	libertarian	2	legal liberty	due process; legal protections; fair trial; probable cause; habeas corpus
free speech press	libertarian-authoritarian	libertarian	3	speech expression	freedom of speech; free speech; free press; freedom of expression; right to criticize government
religious conscience liberty	libertarian-authoritarian	libertarian	3	conscience liberty	religious liberty; freedom of religion; conscience rights; freedom of belief; freedom of worship
assembly protest rights	libertarian-authoritarian	libertarian	3	assembly and protest	right to protest; freedom of assembly; peaceful assembly; right to petition; public dissent
property contract liberty	libertarian-authoritarian	libertarian	3	property contract freedom	secure property rights; freedom of contract; economic liberty; protect property from government
anti surveillance state	libertarian-authoritarian	libertarian	4	anti-surveillance	limit surveillance; end mass surveillance; warrantless surveillance; mass surveillance; surveillance reform
anti police state	libertarian-authoritarian	libertarian	4	anti-coercive policing	police state; limit police powers; police abuse; militarized police; no knock raids
anti executive overreach	libertarian-authoritarian	libertarian	4	anti-executive power	executive overreach; restrain executive power; limit emergency powers; abuse of executive power
anti war anti militarism	libertarian-authoritarian	libertarian	4	anti-militarism	anti war; end foreign wars; bring troops home; non intervention; peaceful foreign policy
personal autonomy	libertarian-authoritarian	libertarian	4	personal choice	personal autonomy; bodily autonomy; victimless crimes; drug decriminalization; end the drug war
radical decentralization	libertarian-authoritarian	libertarian	5	radical localism	radical decentralization; decentralize power; abolish federal control; local sovereignty; decentralized society
abolish coercive state	libertarian-authoritarian	libertarian	5	anti-state coercion	abolish coercive government; abolish state power; end state coercion; abolish prisons; anti state
anarchism	libertarian-authoritarian	libertarian	5	anarchy	anarchy; anarchism; stateless society; no rulers; society without government
libertarian market anarchy	libertarian-authoritarian	libertarian	5	anarcho-capitalism	anarcho capitalism; stateless capitalism; private law; private courts; voluntary market order
libertarian communal anarchy	libertarian-authoritarian	libertarian	5	anarcho-communalism	anarcho communism; libertarian socialism; stateless communism; workers self management; voluntary communes
administrative order	libertarian-authoritarian	authoritarian	1	weak order language	public order; restore order; public safety; enforce the law; law enforcement
national emergency basic	libertarian-authoritarian	authoritarian	1	weak emergency language	national emergency; emergency declaration; state of emergency; emergency powers; extraordinary measures
border enforcement basic	libertarian-authoritarian	authoritarian	2	border control	secure the border; border security; border wall; illegal immigration; immigration enforcement
national security basic	libertarian-authoritarian	authoritarian	2	security state basic	national security; homeland security; security screening; intelligence gathering; terrorist threat

Family	Axis	Direction	Int.	Theme	Representative Variants
traditional morality	libertarian-authoritarian	authoritarian	2	traditionalism	family values; traditional values; sanctity of life; pro life; moral order
law and order	libertarian-authoritarian	authoritarian	3	punitive order	law and order; tough on crime; zero tolerance; restore law and order; strict enforcement
nationalist sovereignty	libertarian-authoritarian	authoritarian	3	nationalism	america first; national sovereignty; protect our sovereignty; national greatness; defend our borders
military strength	libertarian-authoritarian	authoritarian	3	militarism security	military strength; peace through strength; war on terror; military superiority; overwhelming force
surveillance security	libertarian-authoritarian	authoritarian	3	security state	surveillance; security powers; domestic surveillance; watch list; monitor communications
anti protest order	libertarian-authoritarian	authoritarian	3	anti-protest	violent mobs; rioters; domestic extremists; public disorder; riot control
death penalty punitive	libertarian-authoritarian	authoritarian	4	punitive justice	death penalty; capital punishment; execute criminals; maximum penalty; harsh punishment
executive power	libertarian-authoritarian	authoritarian	4	executive dominance	unilateral executive action; strong executive; emergency powers; executive authority; presidential power
military rule emergency	libertarian-authoritarian	authoritarian	4	coercive emergency rule	martial law; military rule; suspend habeas corpus; insurrection act; domestic deployment
anti institutional populism	libertarian-authoritarian	authoritarian	4	anti-institutionalism	deep state; fake news; enemy of the people; rigged system; unelected bureaucrats
mass deportation	libertarian-authoritarian	authoritarian	4	coercive border control	mass deportation; deportation force; remove illegal aliens; deport millions; immigration raids
dictatorial rule	libertarian-authoritarian	authoritarian	5	dictatorship	dictatorship; authoritarian rule; one party rule; absolute power; autocratic rule
suspension of democracy	libertarian-authoritarian	authoritarian	5	democratic breakdown	suspend the constitution; cancel elections; overturn the election; seize power; end democracy
political repression	libertarian-authoritarian	authoritarian	5	state repression	political prisoners; silence dissent; ban opposition; censor the press; jail opponents
extraordinary domestic force	libertarian-authoritarian	authoritarian	5	coercive state power	deploy troops against citizens; shoot protesters; domestic military crackdown; violent suppression; state violence
ethnonationalist authoritarianism	libertarian-authoritarian	authoritarian	5	ethnonationalist extreme	racial purity; blood and soil; white nation; ethno state; white homeland

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